

INSURANCE AI OPERATING MATURITY INDEX

QUESTION DESIGN RATIONALE

P&C Insurance | Group 3 | Week 5 | Mahdin Awsaf Chowdhury — Question Author

The question set is designed to diagnose where a P&C insurer's operating chain breaks rather than how much technology it owns. Every question asks for an observable system state, process, percentage, cadence, or governance practice. Each answer is assigned an explicit value from 1 to 5 and follows the same maturity direction as the Stage 1-5 framework. Percentage bands are diagnostic thresholds, not claimed industry quartiles; they are intentionally non-overlapping so two assessors using the same evidence should select the same level.

Insurance Data Fabric questions test whether the carrier can assemble a trustworthy operating context from policy, claims, billing, CRM, documents, and approved third-party data. The four questions separate identity resolution, refresh cadence, unstructured-document usability, and recurring reporting. This prevents a carrier with a reporting warehouse but weak operational integration from receiving an inflated data score. Pekin's legacy-data transformation and Lemonade's integrated technology stack show that this capability varies materially across real insurers.

Policyholder & Distribution Channels measures more than the existence of a website or app. The questions test digital FNOL share, re-keying, status visibility, and end-to-end journey completion. Those indicators distinguish a digital front door from a genuinely integrated channel. Dryden Mutual's agent-mediated FNOL process provides a low-stage claims-channel anchor, while Lemonade's 96% human-free FNOL demonstrates a materially different operating state.

Claims & Underwriting Automation focuses on straight-through execution rather than isolated task automation. Two questions measure zero-touch claims and endorsement/renewal processing; two test exception routing and whether STP is actively managed as an operating KPI. This matters because a carrier can automate document requests while still relying on manual handoffs throughout the transaction. Fidelity United's pre-modernization manual claims handling, Travelers' measured straight-through eligibility, and Lemonade's roughly 55% end-to-end claim automation provide useful calibration points.

Underwriting & Claims AI separates experimentation from governed production deployment. The questions cover breadth of production use, model inventory/validation, workflow embedding and human authority, and post-deployment monitoring. This is deliberately stricter than asking whether a carrier “uses AI.” ProAssurance and Westfield illustrate low/early maturity, while Lemonade demonstrates deeply embedded operational AI. The NAIC Model Bulletin supplies the governance expectation for the upper stages; because public disclosures rarely prove every governance control, the rationale does not overstate any company as fully Stage 5 on governance.

Operating Change Capacity measures whether approved decisions become deployed, adopted, measured change at business speed. Cycle time, standing cross-functional ownership, post-release measurement, and pre-defined accountability are observable proxies for execution capacity. Pekin's historical 18-month product introduction cycle and Texas Mutual's two-week delivery cadence create a clear low-to-high calibration. Together, the 20 questions should produce visibly different Stage-2 and Stage-4 radar patterns and give the tool deterministic 1-5 inputs.

DESIGN GUARDRAILS

Observable	Monotonic	Calibrated, not overstated
Questions score evidence, not self-perception.	Every answer maps explicitly from 1 to 5 in the same maturity direction.	Company examples anchor specific capabilities; they are not blanket maturity labels.

SELECTED EVIDENCE REGISTER

- Pekin Insurance — ValueMomentum data modernization case study; Novarica/Research Council case study compendium (2019).
- Lemonade — Form 10-K for fiscal year 2025 (SEC).
- Dryden Mutual — Claims process.
- Fidelity United Insurance — Hyland customer case study.
- Travelers — 2025 Annual Report.
- ProAssurance — “AI and Medical Liability” (Feb. 2024).
- Westfield — Claims AI implementation interview (Insurance Journal, Apr. 8, 2025).
- Texas Mutual — Agile Velocity transformation case study.
- NAIC — Model Bulletin on the Use of Artificial Intelligence Systems by Insurers (Dec. 4, 2023).